Internship Report

on

SmartData AI Agent: A Tool-Using Assistant for Data Analysis and Prediction

Submitted by

Rutuja Devanand Soundale

Roll No. EN23102062

B.Tech Electronics and Telecommunication Engineering

In partial fulfilment of the requirements for the degree of

Bachelor of Technology

in

Electronics and Telecommunication Engineering

Under the guidance of

Dr. Parshuram Arotale Faculty Guide

Dr. Alok Yadav Industry Guide

YBI Foundation

DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING

VIDYA PRATISHTHAN'S

KAMALNAYAN BAJAJ INSTITUTE OF ENGINEERING AND TECHNOLOGY, BARAMATI

2025 - 2026

CERTIFICATE

This is to certify that **RUTUJA DEVANAND SOUNDALE**, Roll No. EN23102062, a student of Bachelor of Technology in Electronics and Telecommunication Engineering at Vidya Pratishthan's Kamalnayan Bajaj Institute of Engineering and Technology, Baramati, has successfully completed the internship at **YBI FOUNDATION** in the domain of Artificial Intelligence and Generative AI during the period from 18th June 2026 to 18th October 2026.

The internship work included Python programming, data analysis and visualization, machine learning and predictive modelling, deep learning with artificial neural networks, binary and multi-class classification, an introduction to Generative AI, and the design and development of a basic tool-using AI agent for data analysis and prediction.

The work presented in this report was carried out during the internship period under the guidance and supervision of the industry guide and the faculty guide.

Dr. Alok Yadav
Industry Guide

Dr. Parshuram Arotale
Faculty Guide

External Examiner

Dr. S. B. Lande
Principal

Head of Department
Department of E&TC

Date:

ACKNOWLEDGEMENT

I am thankful to everyone who supported me during my internship and during the preparation of this report.

I sincerely thank Dr. Alok Yadav, my mentor and industry guide at YBI Foundation, for the structured weekly training, for the regular review meetings and for the valuable suggestions that helped me improve my work. I also thank YBI Foundation for giving me the opportunity to learn Artificial Intelligence and Generative AI through practical, project-based training.

I am grateful to my faculty guide, Dr. Parshuram Arotale, for the guidance and encouragement given during the internship and in the preparation of this report.

I express my thanks to the Head of the Department of Electronics and Telecommunication Engineering and to the faculty members of the department for their support. I am also thankful to the Principal, Dr. S. B. Lande, and to Vidya Pratishthan's Kamalnayan Bajaj Institute of Engineering and Technology, Baramati, for providing an environment that encourages internships and practical learning.

Finally, I thank my family and friends for their constant support and motivation.

Rutuja Devanand Soundale

ABSTRACT

This report describes my four-month internship in Artificial Intelligence and Generative AI at YBI Foundation from 18 June 2026 to 18 October 2026, carried out under the mentorship of Dr. Alok Yadav. The internship followed a structured curriculum. It started with Python programming: setup, basics, operators, conditional statements, loops and functions. It continued with data analysis using Pandas: reading and writing external data, data summarization, data manipulation and building data dashboards with Matplotlib.

The next phase covered predictive modelling and predictive analytics. I studied and applied Logistic Regression, K-Nearest Neighbours (KNN), Decision Trees, Random Forest, K-Means and clustering models to business problems. This was followed by deep learning, Artificial Neural Network (ANN) regression and binary and multi-class classification, along with their advantages, disadvantages, applications and limitations. I was also introduced to Generative AI, large language models and prompt engineering.

In the final phase I designed and built the SmartData AI Agent, a basic tool-using AI agent written in Python. A user types a request in plain English, and the agent follows a simple Perceive, Think, Act, Observe and Respond loop. Its brain decides which of six tools to use: dataset summary, chart creation, model comparison, species prediction, K-Means clustering and a safe calculator. The brain is rule-based and works fully offline, and it can optionally use the Google Gemini large language model to choose tools. The agent also keeps a short conversation memory.

The agent was tested with eleven test cases covering every tool and also unclear or unsupported requests, and it handled all of them correctly. On the Iris dataset, the four models reached test accuracies of 94.7% (Logistic Regression), 97.4% (KNN), 89.5% (Decision Tree) and 92.1% (Random Forest), and the agent automatically selected KNN for prediction. The report also presents the week-wise work, the skills gained, the challenges faced, the limitations of the project and its future scope.

Keywords: Python; Pandas; data visualization; machine learning; logistic regression; KNN; decision tree; random forest; K-Means; artificial neural network; classification; Generative AI; large language model; AI agent; tool calling.

DECLARATION

I, Rutuja Devanand Soundale, Roll No. EN23102062, a student of Bachelor of Technology in Electronics and Telecommunication Engineering at Vidya Pratishthan's Kamalnayan Bajaj Institute of Engineering and Technology, Baramati, declare that this internship report titled "SmartData AI Agent: A Tool-Using Assistant for Data Analysis and Prediction" is the result of my own work carried out during my internship at YBI Foundation from 18th June 2026 to 18th October 2026.

Ideas, figures and text taken from other sources are acknowledged through citations and listed in the References. This report does not contain any confidential data, passwords or API keys.

Rutuja Devanand Soundale
Roll No. EN23102062

Date:
Place: Baramati

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CHAPTER 1 INTRODUCTION

1.1 Internship Overview

I completed a four-month live project-based internship in Artificial Intelligence and Generative AI at YBI Foundation. The internship started on 18 June 2026 and was completed on 18 October 2026, a period of about 18 weeks (120 days). The internship was conducted online and was unpaid. My mentor and industry guide was Dr. Alok Yadav.

The internship followed a clear path. In the first weeks I learned Python programming. I then learned how to read, summarize, manipulate and visualize data with Pandas and Matplotlib. After that I studied machine learning models used for predictive modelling and predictive analytics, and applied each of them to business problems. This was followed by deep learning, Artificial Neural Networks (ANN) and classification models, together with an introduction to Generative AI. In the last five weeks I used this knowledge to design and build a project called the SmartData AI Agent.

Every week ended with a weekly internship report issued by YBI Foundation that listed the daily tasks. These tasks are summarized in Chapter 10. Regular discussions with the mentor guided the project work in the final weeks.

1.2 Objectives of the Internship

The objectives of the internship were:

- to learn Python programming from setup to functions and use it to solve problems;
- to read, summarize, manipulate and visualize data using Pandas and Matplotlib;
- to understand supervised and unsupervised machine learning models and apply them to business problems;
- to understand deep learning, ANN regression, and binary and multi-class classification, including their advantages, disadvantages and limitations;
- to understand the basics of Generative AI, large language models (LLMs) and prompt engineering;
- to design, build, test and document a basic AI agent that brings these skills together in one working application.

1.3 Scope of the Internship

The learning part of the internship covered the full path from Python basics to machine learning, deep learning and Generative AI. The project part is a basic AI agent that works on the Iris flower dataset. The dataset is small and well known, so the focus stays on how an agent works: how it understands a request, how it chooses a tool, how the tool does the work and how the answer is returned. The agent runs in the command line on a normal laptop and does not need the internet. The optional LLM brain needs an internet connection and an API key.

1.4 Technologies and Platforms Used

Table 1: Technologies and platforms used

Category	Technology / Platform	Use in the internship
Language	Python 3	All programs, exercises and the project
Data handling	NumPy, Pandas	Reading, summarizing and manipulating data
Visualization	Matplotlib	Charts and data dashboards
Machine learning	scikit-learn	Logistic Regression, KNN, Decision Tree, Random Forest, K-Means
Deep learning	Artificial Neural Networks (Keras / TensorFlow)	ANN regression and classification exercises
Generative AI	Google Gemini, ChatGPT	Prompt engineering; optional LLM brain of the agent
Notebooks	Jupyter Notebook, Google Colab	Practice and experiments
Editor and version control	VS Code, Git, GitHub	Writing the project code and storing it

1.5 Methodology

The internship was divided into five phases. Each phase built on the previous one. For every topic the order of work was the same: understand the concept, study its application to a business problem, and then implement it in Python. Figure 1 shows the phases.

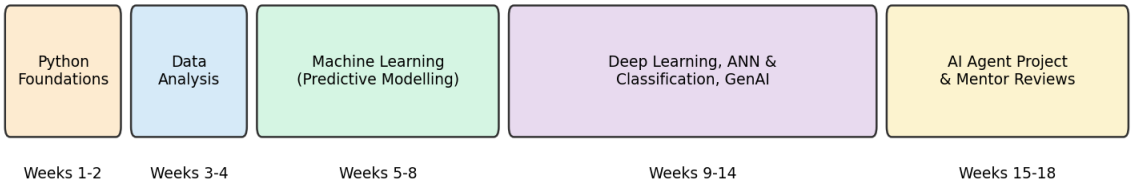


Figure 1: Phases of the internship

1.6 Organization of the Report

Chapter 2 describes YBI Foundation and the way the internship was carried out. Chapters 3 to 6 explain what I learned: Python, data analysis, machine learning, and deep learning with Generative AI. Chapter 7 presents the design of the SmartData AI Agent and Chapter 8 its implementation. Chapter 9 gives the testing and results. Chapter 10 gives the week-wise summary, Chapter 11 the skills, challenges and learning outcomes, and Chapter 12 the conclusion and future scope.

CHAPTER 2 ORGANIZATION AND INTERNSHIP BACKGROUND

2.1 Organization Overview

YBI Foundation is an industry-oriented training and internship provider in the areas of Artificial Intelligence, Machine Learning, Data Science, Python and Generative AI. Its internships follow a hands-on, project-based learning model, and students receive a certificate on successful completion. The internship is delivered online, with learning modules, weekly reports and mentor guidance.

2.2 My Role and Way of Working

I joined as an intern in the Artificial Intelligence and Generative AI domain on 18 June 2026. Each week had five working days, and each day had a defined task. I studied the topic of the day, practised it in a notebook and applied it to an example problem. At the end of every week YBI Foundation issued a weekly internship report listing the tasks of that week, signed by the HR Manager. In the final phase, project work was combined with regular discussions with my mentor, Dr. Alok Yadav, and I incorporated the mentor's suggestions into the project.

2.3 Internship Curriculum

The curriculum followed by YBI Foundation is shown in Table 2. A first review presentation was given after about one and a half months of the internship, when the Python, Python libraries and data visualization modules were complete and the machine learning module was in progress.

Table 2: Internship curriculum of YBI Foundation

Module	Topic	Main contents
1	Introduction	Orientation, overview of AI and Generative AI, tool setup
2	Python Programming	Setup, basics, operators, conditional statements, loops, functions
3	Python Libraries	NumPy and Pandas: reading, writing, summarizing and manipulating data
4	Data Visualization	Matplotlib charts and data dashboards
5	Machine Learning	Regression, classification, KNN, decision trees, random forest, K-Means, clustering

Module	Topic	Main contents
6	Artificial Intelligence	Deep learning, ANN regression, binary and multi-class classification
7	Generative AI	LLMs, prompt engineering, hands-on with ChatGPT and Gemini, responsible AI
8	Project Work	Design and development of the project with mentor discussions
9	Project Submission	Testing, documentation, final review

2.4 Why an AI Agent Project

Most of the training covered separate skills: Python, Pandas, charts and several machine learning models. A real AI application must combine these skills. An AI agent is a good way to do this, because it is a program that understands a request, decides what to do, uses tools to do the work and returns an answer. Each tool of my agent uses one skill from the internship, so the project brings the whole internship together in one working application that is also easy to demonstrate and explain.

CHAPTER 3 PYTHON PROGRAMMING FOUNDATIONS

Python was the base of the whole internship. It is simple to read, has a very large set of libraries for data science and AI, and is the most common language used for machine learning. Weeks 1 to 3 covered the topics below.

3.1 Python Setup and Installation

I installed Python, set up Jupyter Notebook and used Google Colab, which runs Python in the browser without any installation. I learned to install libraries with pip and to run programs both in notebooks and from the command line.

3.2 Python Basics

I learned variables, data types (int, float, string, boolean), type conversion, input and output, and the main data structures: lists, tuples, sets and dictionaries. Dictionaries were especially useful later, because the agent stores its tools in a dictionary that maps a tool name to a function.

3.3 Python Operators

I studied arithmetic operators (+, -, *, /, //, %, **), comparison operators, logical operators (and, or, not), assignment operators and membership operators (in, not in). The calculator tool of the agent supports the arithmetic operators, and the brain of the agent uses the membership operator to check whether a keyword is present in the user's message.

3.4 Conditional Statements

I learned if, elif and else statements to make decisions in a program. The rule-based brain of the agent is built from a chain of such conditions: if the message contains the word "predict" and four numbers, the prediction tool is chosen; otherwise the next condition is checked.

3.5 Loops

I learned for loops, while loops, the range() function, and the break and continue statements. The main program of the agent is a while loop that keeps reading messages until the user types exit.

3.6 Functions

I learned how to define functions with def, pass parameters, use default values and return results. Functions make code reusable and easy to test. In the project, every tool is a separate Python function, which keeps the code simple and easy to explain.

```
def calculator(expression):
    result = evaluate(expression)
    return f"{expression} = {result}"
```

CHAPTER 4 DATA ANALYSIS AND VISUALIZATION

Weeks 3 and 4 covered data analysis with the Pandas library and data visualization with Matplotlib. These skills are needed before any machine learning, because a model is only as good as the data given to it.

4.1 External Data Read and Write

I learned to read data from CSV and Excel files into a Pandas DataFrame with read_csv() and read_excel(), and to save results back with to_csv(). A DataFrame is a table with rows and named columns, similar to a spreadsheet.

4.2 Data Summarization

I used head(), info(), shape, describe(), value_counts() and groupby() to understand a dataset quickly: how many rows and columns it has, what type each column is, the average and spread of each numeric column, and how many records belong to each category. The dataset summary tool of the agent uses exactly these functions.

4.3 Data Manipulation

I learned to select columns and rows, filter rows with conditions, sort data, add new columns, handle missing values with isnull(), fillna() and dropna(), rename columns, and merge or join tables. These steps prepare raw data for analysis and modelling.

4.4 Data Visualization and Dashboards

I used Matplotlib to make line charts, bar charts, histograms, scatter plots and pie charts, and combined several charts in one figure with subplots to build a simple data dashboard. A good chart shows patterns that are hard to see in a table. For example, the scatter plot in Figure 2 shows that the three Iris species form clear groups by petal size; this was produced by the chart tool of the agent.

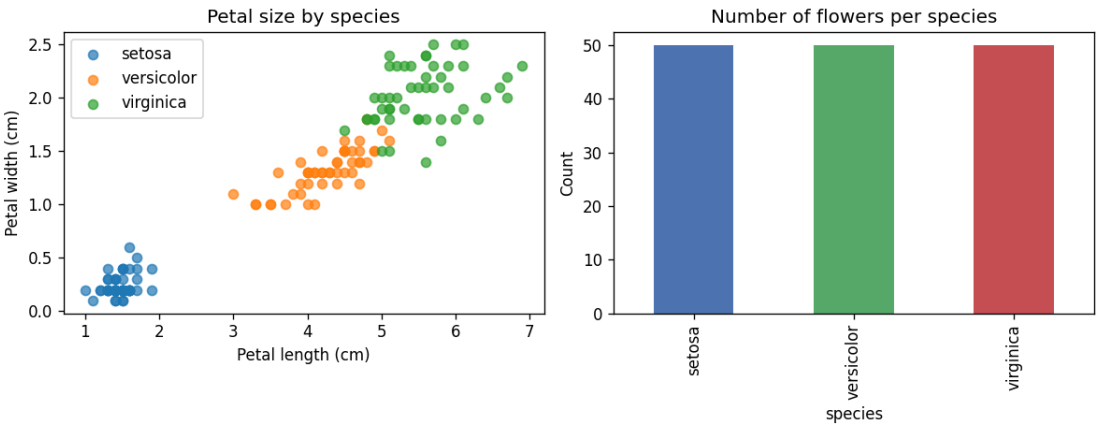


Figure 2: Data dashboard of the Iris dataset produced by the agent's chart tool

CHAPTER 5 MACHINE LEARNING AND PREDICTIVE MODELLING

Weeks 4 to 8 covered predictive modelling and predictive analytics. Machine learning is a way to let a computer learn patterns from past data and use them to make predictions on new data. For each model I first understood how it works, then studied its application to a business problem, and then implemented it in Python with scikit-learn.

5.1 Types of Machine Learning

- **Supervised learning:** the data has a known answer (label). The model learns the link between inputs and the label. Regression predicts a number (for example, sales); classification predicts a category (for example, whether a customer will leave).
- **Unsupervised learning:** the data has no label. The model finds hidden groups or patterns by itself, for example customer segments.

5.2 Standard Workflow for a Business Problem

Every model was implemented with the same steps: define the business problem; collect and clean the data; choose the input features (X) and the target (y); split the data into a training set and a test set with train_test_split(); train the model with fit(); make predictions with predict(); and evaluate the model using accuracy, a confusion matrix, precision and recall, or error measures for regression. This workflow is also used inside the model comparison tool of the agent.

5.3 Logistic Regression

Logistic Regression is a classification model. It calculates a weighted sum of the inputs and passes it through the sigmoid function, which gives a probability between 0 and 1. If the probability is above a threshold, the record is put in the positive class. Business applications include predicting customer churn, loan default and whether an email is spam. It is fast and easy to interpret, but it works best when classes can be separated by a straight boundary.

5.4 K-Nearest Neighbours (KNN)

KNN classifies a new record by looking at the K closest records in the training data and taking a majority vote. It needs no real training step and is easy to understand. Its applications include recommendation systems and customer classification. Its drawbacks are that it becomes slow on large datasets and it needs features on a similar scale.

5.5 Decision Trees

A Decision Tree asks a series of yes/no questions about the features, such as "Is petal length less than 2.5 cm?", and follows the branches to reach a decision. It is easy to visualize and explain to non-technical people, which makes it useful for credit approval and customer segmentation. A single tree can overfit, which means it memorizes the training data and performs worse on new data.

5.6 Random Forest

Random Forest builds many decision trees on random samples of the data and features, and combines their votes. This reduces overfitting and usually gives better and more stable accuracy than a single tree. It is used for fraud detection, sales forecasting and risk scoring. It is slower and harder to interpret than a single tree.

5.7 K-Means and Clustering

K-Means is an unsupervised model. It places K centre points, assigns every record to the nearest centre, moves each centre to the mean of its records, and repeats until the groups stop changing. Clustering is used for customer segmentation, market analysis and grouping similar products. The number of groups K must be chosen in advance, for example with the elbow method.

Table 3: Comparison of the machine learning models studied

Model	Type	Advantage	Limitation
Logistic Regression	Supervised, classification	Fast, gives probabilities, easy to interpret	Assumes a simple (linear) boundary
KNN	Supervised, classification	Simple, no training step	Slow on large data; needs scaling
Decision Tree	Supervised, classification / regression	Easy to visualize and explain	Can overfit
Random Forest	Supervised, classification / regression	High accuracy, robust	Slower, less interpretable
K-Means	Unsupervised, clustering	Finds groups without labels	K must be chosen; sensitive to scale

5.8 Review and Feedback

At the end of Week 8 the work done so far was reviewed and I incorporated the mentor's suggestions, mainly about clearly explaining the business problem before building a model and always checking the model on unseen test data.

CHAPTER 6 DEEP LEARNING, CLASSIFICATION AND GENERATIVE AI

Weeks 9 to 14 covered deep learning, artificial neural networks and classification models, together with an introduction to Generative AI. For each model family I studied the understanding, advantages, disadvantages, applications and limitations, and implemented it on a business problem.

6.1 Deep Learning

Deep learning is a part of machine learning that uses neural networks with many layers. It performs very well on large and complex data such as images, speech and text, because it learns useful features automatically. I also studied its limitations: it needs a large amount of data, high computing power and long training time, and it behaves like a black box that is hard to explain. For small tabular datasets, simple models such as Random Forest are often just as good.

6.2 Artificial Neural Networks for Regression

An Artificial Neural Network (ANN) is made of an input layer, one or more hidden layers and an output layer. Each neuron calculates a weighted sum of its inputs and applies an activation function such as ReLU. The network learns by comparing its output with the correct answer using a loss function and adjusting the weights through backpropagation and gradient descent. For regression, the output layer has a single neuron without activation and the loss is the mean squared error. Advantages are the ability to model non-linear relationships; disadvantages are the need for more data, tuning of many settings and the risk of overfitting.

6.3 Binary Classification

Binary classification predicts one of two classes, for example whether a customer will buy or not. In an ANN, the output layer has one neuron with a sigmoid activation and the loss is binary cross-entropy. The model is evaluated with accuracy, precision, recall, F1-score and the confusion matrix.

6.4 Multi-Class Classification

Multi-class classification predicts one of three or more classes, such as the three Iris species used in my project. In an ANN, the output layer has one neuron per class with a softmax activation, which gives a probability for each class, and the loss is categorical cross-

entropy. Applications include product categorization, handwritten digit recognition and document classification. Its limitations are that it needs enough examples of every class and that similar classes are easily confused.

Table 4: Summary of deep learning and classification topics

Topic	Advantages	Disadvantages / Limitations	Business application
Deep learning	Learns features automatically; best for images, text, speech	Needs large data and computing power; hard to explain	Image and speech recognition
ANN regression	Models non-linear relationships	Needs tuning; can overfit	Price and demand prediction
Binary classification	Simple output; clear metrics	Class imbalance affects results	Churn and fraud detection
Multi-class classification	Handles many categories at once	Similar classes get confused	Product and document categorization

6.5 Generative AI and Large Language Models

Generative AI creates new content such as text, images or code. A Large Language Model (LLM), such as ChatGPT or Google Gemini, is trained on a very large amount of text and generates an answer one word piece (token) at a time. I learned prompt engineering: writing clear instructions, giving examples, asking for a fixed output format such as JSON, and checking the answer, because an LLM can produce confident but wrong answers. I also studied responsible use of AI: privacy, bias and the need for human checking.

6.6 From Models to Agents

A model answers one fixed kind of question. An AI agent is a program that can understand a goal, decide which action to take, use tools to take that action, and look at the result before answering. Modern agents often use an LLM as the brain that chooses the tools, while normal code performs the actual work. This idea became the basis of my project.

CHAPTER 7 PROJECT DESIGN: SMARTDATA AI AGENT

7.1 Problem Statement

A beginner who wants to analyze a dataset must remember many different commands: one for a summary, another for a chart, several for training and comparing models, and more for predictions. The goal of this project is to build a basic AI agent that accepts a request in simple English, decides by itself which analysis is needed, performs it and explains the result.

7.2 Project Objectives

- to build an agent that understands simple English requests about a dataset;
- to give the agent tools that use the skills learned in the internship: Pandas summary, Matplotlib charts, four classification models, K-Means clustering and a calculator;
- to make the agent choose the correct tool automatically, using a rule-based brain that works offline and an optional LLM brain;
- to keep a short memory of the conversation;
- to handle unclear or unsupported requests safely instead of crashing or guessing;
- to keep the code small and clear so that it is easy to explain.

7.3 What Is an AI Agent? (Simple Explanation)

A simple way to explain the project is to compare the agent with a helpful assistant in an office. The assistant listens to the request (perceive), thinks about which tool or person can do the job (think), uses that tool (act), checks the result (observe) and then reports back (respond). The assistant also remembers what was asked earlier (memory). The SmartData AI Agent works in exactly this way, as shown in Figure 3.

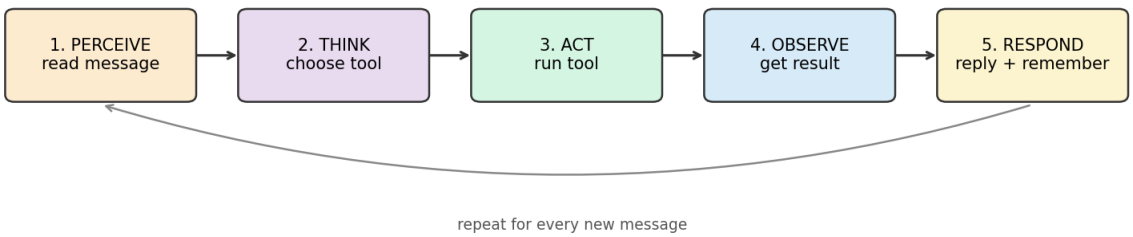


Figure 3: The Perceive-Think-Act-Observe-Respond loop of the agent

7.4 System Architecture

The agent has three parts, each in its own Python file (Figure 4):

- **agent.py (the agent loop):** receives the message, asks the brain for a decision, runs the chosen tool, stores the result in memory and prints the answer.
- **brain.py (the brain):** decides which tool to use and with which inputs. It returns a decision such as {"tool": "predict_species", "args": {...}}.
- **tools.py (the hands):** contains the six tools. Each tool is a normal Python function that does one job.

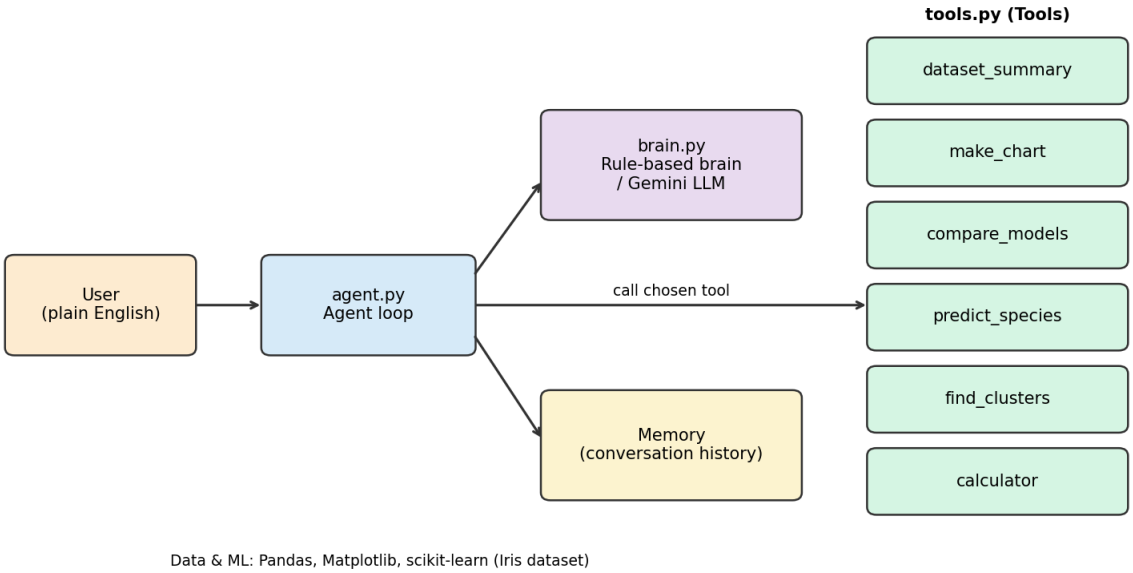


Figure 4: Architecture of the SmartData AI Agent

7.5 Tools of the Agent

Table 5: Tools of the agent and the learning behind each tool

Tool	What it does	Internship learning used
dataset_summary	Rows, columns, flowers per species, average measurements	Pandas: summarization, groupby (Week 3)
make_chart	Scatter plot and bar chart saved as an image	Matplotlib dashboards (Weeks 3-4)
compare_models	Trains Logistic Regression, KNN, Decision Tree, Random Forest and reports accuracy	Predictive modelling (Weeks 5-7)
predict_species	Predicts the species of a new flower with the best model	Multi-class classification (Weeks 13-14)
find_clusters	Groups flowers with K-Means and compares groups with real species	K-Means and clustering (Weeks 7-8)
calculator	Safely evaluates arithmetic such as (45+15)*2/3	Python operators and functions (Weeks 2-3)

7.6 The Brain: Rule-Based and LLM

The rule-based brain reads the message in lower case, finds numbers with a regular expression, and checks keywords in a fixed order. For example, a message with "predict" and four numbers goes to predict_species, and a message with "compare" or "accuracy" goes to compare_models. If nothing matches, the agent politely says that it did not understand and suggests typing help. This brain works without the internet and always gives the same result for the same input.

The optional LLM brain sends the user's message and the list of tools to Google Gemini and asks it to reply only in JSON with the tool name and its inputs. The LLM only chooses the tool; the calculation is still done by the Python tool, so the numbers in the answer are always correct. If the LLM fails, returns invalid JSON or names a tool that does not exist, the agent automatically falls back to the rule-based brain. This acts as a simple guardrail.

7.7 Dataset

The project uses the Iris flower dataset, which is built into scikit-learn. It contains 150 flowers of three species (setosa, versicolor and virginica), 50 of each, with four measurements in centimetres: sepal length, sepal width, petal length and petal width. It is clean, balanced and easy to understand, which makes it ideal for explaining multi-class classification and clustering. The data is split into 112 training flowers and 38 test flowers (75% / 25%), keeping the same species ratio in both sets.

CHAPTER 8 IMPLEMENTATION

8.1 Project Structure

```
ai-agent/
  agent.py          - agent loop and command-line chat (run this file)
  brain.py          - rule-based brain and optional Gemini LLM brain
  tools.py          - the six tools and the tool registry
  requirements.txt  - pandas, numpy, matplotlib, scikit-learn
  README.md         - how to run the project
  outputs/          - saved chart and sample run
```

8.2 The Tool Registry

All tools are listed in one dictionary. The key is the tool name and the value is the function with a short description. The brain only needs this list to know what the agent can do, and a new tool can be added with a single line.

```
TOOLS = {
    "dataset_summary": (dataset_summary, "Describe the Iris dataset"),
    "make_chart":      (make_chart,      "Draw and save a chart"),
    "compare_models":  (compare_models,  "Train and compare 4 ML models"),
    "predict_species": (predict_species, "Predict species from 4 measurements"),
    "find_clusters":   (find_clusters,   "Group flowers using K-Means"),
    "calculator":      (calculator,      "Evaluate an arithmetic expression"),
}
```

8.3 Model Comparison Tool

This tool trains the four models on the same training data, measures their accuracy on the same test data and remembers the best one. The prediction tool then uses that best model.

```
models = {
    "Logistic Regression": LogisticRegression(max_iter=500),
    "KNN (k=5)":           KNeighborsClassifier(n_neighbors=5),
    "Decision Tree":       DecisionTreeClassifier(random_state=42),
    "Random Forest":       RandomForestClassifier(n_estimators=100, random_state=42),
}
for name, model in models.items():
    model.fit(X_train, y_train)
    scores[name] = accuracy_score(y_test, model.predict(X_test))
best_model_name = max(scores, key=scores.get)
```

8.4 The Rule-Based Brain

```
def decide(self, message):
    text = message.lower()
    numbers = [float(n) for n in re.findall(NUMBER, text)]
    if "predict" in text and len(numbers) >= 4:
        return {"tool": "predict_species", "args": {...first four numbers...}}
    if "compare" in text or "accuracy" in text:
        return {"tool": "compare_models", "args": {}}
    ...
    return {"tool": None, "reply": "I did not understand that. Type 'help'."}
```

8.5 The Agent Loop

The handle() method of the agent is the heart of the project. It follows the five steps of the agent loop and also catches errors from a tool, so the agent never crashes on a bad input.

```
def handle(self, message):
    decision = self.brain.decide(message)            # THINK
    tool_name = decision.get("tool")
    if tool_name is None:
        answer = decision.get("reply")
    else:
        function, _ = TOOLS[tool_name]
        try:
            answer = function(**decision.get("args", {}))   # ACT + OBSERVE
        except Exception as error:
            answer = f"The tool '{tool_name}' failed: {error}"
    self.memory.append((message, tool_name, answer))     # REMEMBER
    return answer                                        # RESPOND
```

8.6 Safe Calculator

Python's eval() function can run any code, which is unsafe if a user types something harmful. The calculator therefore uses Python's ast module to read the expression and allows only numbers and the operators +, -, *, /, ** and %. Anything else is rejected with a clear message.

8.7 Running the Project

```
pip install -r requirements.txt
python agent.py --offline      # rule-based brain, no internet needed
python agent.py                # uses Gemini if GEMINI_API_KEY is set
```

CHAPTER 9 TESTING, RESULTS AND DISCUSSION

9.1 Testing Approach

The agent was tested by giving it a fixed list of eleven messages in the offline mode, so that the results can be repeated exactly. The messages cover every tool, two different ways of asking for a prediction, a prediction request with missing numbers, an unsupported question and the memory command. The complete output is saved in outputs/demo_run.txt and reproduced in Appendix A.

9.2 Functional Test Cases

Table 6: Functional test cases of the agent (offline mode)

No.	User message	Tool chosen	Result	Status
1	help	show_help	Listed the seven things the agent can do	Pass
2	summary of the dataset	dataset_summary	150 rows, 4 inputs, 50 flowers per species, averages per species	Pass
3	draw a chart	make_chart	Dashboard image saved (Figure 2)	Pass
4	compare models	compare_models	Accuracy of 4 models; best model KNN	Pass
5	predict 5.1 3.5 1.4 0.2	predict_species	setosa (confidence 100%)	Pass
6	which species is 6.7 3.0 5.2 2.3	predict_species	virginica (confidence 100%)	Pass
7	predict this flower	none	Asked the user for the 4 measurements	Pass
8	find clusters	find_clusters	3 groups; setosa perfectly separated	Pass
9	calculate (45+15)*2/3	calculator	40.0	Pass
10	what is the weather	none	Polite message suggesting 'help'	Pass
11	history	memory	Listed the 10 earlier messages and tools	Pass

All eleven test cases passed. Cases 7 and 10 are important: when the request is incomplete or outside the agent's abilities, the agent does not guess or crash but explains what it needs.

9.3 Model Comparison Results

Table 7: Accuracy of the four models on the test set

Model	Test accuracy	Correct out of 38
Logistic Regression	94.7%	36
KNN (k = 5)	97.4%	37
Decision Tree	89.5%	34
Random Forest	92.1%	35

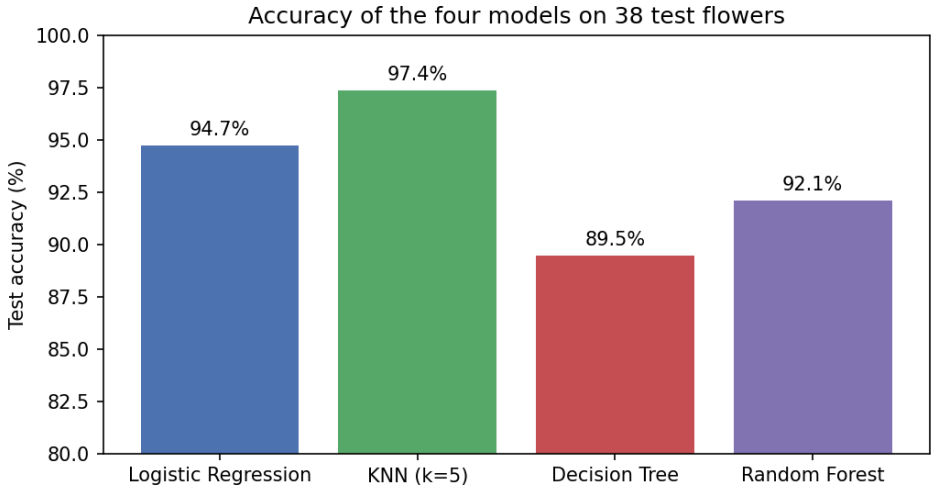


Figure 5: Test accuracy of the four classification models

KNN gave the best accuracy (97.4%) and was selected automatically by the agent for predictions. Figure 6 shows its confusion matrix: all setosa and versicolor test flowers were classified correctly and only one virginica flower was confused with versicolor. These two species have overlapping measurements, as is also visible in the scatter plot of Figure 2.

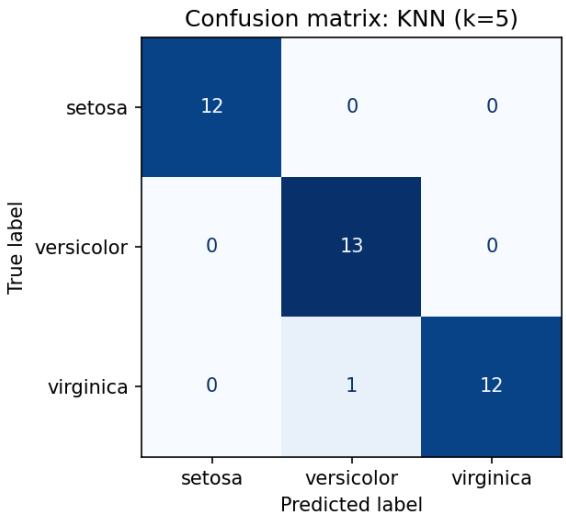


Figure 6: Confusion matrix of the best model (KNN) on 38 test flowers

9.4 Clustering Result

Table 8: K-Means groups compared with the real species

K-Means group	setosa	versicolor	virginica
Group 0	0	48	14
Group 1	50	0	0
Group 2	0	2	36

Without being given any labels, K-Means placed all 50 setosa flowers in one group. 48 of 50 versicolor and 36 of 50 virginica flowers were placed in their own groups, so 134 of the 150 flowers (89.3%) fall in a group that matches their species. This shows how unsupervised learning can discover natural groups, and also that it is less accurate than supervised learning when classes overlap.

9.5 Discussion

The results show that a small amount of clear code is enough to build a useful agent. Separating the brain from the tools has two benefits. First, the brain can be changed from rules to an LLM without touching the tools. Second, because the tools do the real calculations, the numbers in the answers are always exact, even when an LLM is used to choose the tool. The fallback from the LLM brain to the rule-based brain keeps the agent working even without an internet connection.

9.6 Limitations

- The rule-based brain understands only the keywords it was written for; unusual wording may not be understood.
- The agent works on one built-in dataset; it cannot yet load any CSV file chosen by the user.
- The test set is small (38 flowers), so a difference of one flower changes the accuracy by about 2.6%.
- The memory lasts only while the program is running.
- The LLM brain needs an internet connection and an API key, and its tool choice can vary between runs.

CHAPTER 10 WEEK-WISE INTERNSHIP SUMMARY

10.1 Overview

This chapter lists the tasks of each week as recorded in the weekly internship reports issued by YBI Foundation. Each week had five working days (Monday to Friday).

10.2 Phase 1: Python Foundations (Weeks 1-2)

Table 9: Weeks 1-2: Python foundations

Week	Dates	Tasks and learning
1	18-06-2026 to 19-06-2026	Joining and orientation; introduction to the internship curriculum, Artificial Intelligence and Generative AI; setting up the learning tools
2	22-06-2026 to 26-06-2026	Python Training: Python setup and installation; Python basics; Python operators; Python conditional statements; Python loops

10.3 Phase 2: Data Analysis and Visualization (Weeks 3-4)

Table 10: Weeks 3-4: data analysis and visualization

Week	Dates	Tasks and learning
3	29-06-2026 to 03-07-2026	Python Training: Python functions. Data Analysis: external data read and write; data summarization; data manipulation; data dashboard
4	06-07-2026 to 10-07-2026	Data visualization practice; introduction to machine learning and predictive modelling; understanding regression models and applying them to business problems

10.4 Phase 3: Machine Learning (Weeks 5-8)

Table 11: Weeks 5-8: machine learning and predictive analytics

Week	Dates	Tasks and learning
5	13-07-2026 to 17-07-2026	Predictive Modelling: implementation to solve business problems; understanding Logistic Regression; application of Logistic Regression for business; implementation to solve business problems; understanding the KNN model
6	20-07-2026 to 24-07-2026	Predictive Modelling: application of the KNN model for business; implementation to solve business problems. Predictive Analytics: understanding the Decision Trees model; application of Decision Trees for business; implementation to solve business problems

Week	Dates	Tasks and learning
7	27-07-2026 to 31-07-2026	Predictive Analytics: understanding the Random Forest model; application of Random Forest for business; implementation to solve business problems; understanding the K-Means model; application of K-Means for business
8	03-08-2026 to 07-08-2026	Predictive Analytics: implementation to solve business problems; understanding the clustering model; application of the clustering model for business; implementation to solve business problems. Review and feedback: incorporating mentor suggestions

10.5 Phase 4: Deep Learning, Classification and Generative AI (Weeks 9-14)

Table 12: Weeks 9-14: deep learning, classification and Generative AI

Week	Dates	Tasks and learning
9	10-08-2026 to 14-08-2026	Introduction to Generative AI and large language models; prompt engineering; hands-on with ChatGPT and Gemini; responsible use of AI
10	17-08-2026 to 21-08-2026	Deep Learning: understanding models; advantages; disadvantages; applications of deep learning models
11	24-08-2026 to 28-08-2026	Deep Learning: limitations of models; implementation to solve business problems. ANN Regression: understanding models; advantages of models; disadvantages of models
12	31-08-2026 to 04-09-2026	ANN Regression: applications; limitations; implementation to solve business problems. Binary Classification: understanding models
13	07-09-2026 to 11-09-2026	Binary Classification: advantages; disadvantages; applications and limitations; implementation to solve business problems. Multi Classification: understanding models
14	14-09-2026 to 18-09-2026	Multi Classification: advantages of models; disadvantages of models; applications of models; limitations of models; implementation to solve business problems

10.6 Phase 5: AI Agent Project (Weeks 15-18)

Table 13: Weeks 15-18: AI agent project

Week	Dates	Tasks and learning
15	21-09-2026 to 25-09-2026	Working on project (Mon, Tue, Wed, Fri); discussion with mentor (Thu). Selected the AI agent idea and the Iris dataset; designed the agent loop and the tool list

Week	Dates	Tasks and learning
16	28-09-2026 to 02-10-2026	Working on project (Mon, Tue, Thu, Fri); discussion with mentor (Wed). Built the tools: summary, chart, model comparison, prediction, clustering, calculator
17	05-10-2026 to 09-10-2026	Working on project (Mon, Wed, Thu, Fri); discussion with mentor (Tue). Built the rule-based brain, the optional Gemini brain with fallback, and the conversation memory
18	12-10-2026 to 15-10-2026	Discussion with mentor (Mon); working on project (Tue-Thu). Testing, fixing issues, documentation and final review. Performance review on 16-10-2026

CHAPTER 11 SKILLS, CHALLENGES AND LEARNING OUTCOMES

11.1 Technical Skills Gained

Table 14: Technical skills gained during the internship

Area	Skills
Python	Data types, operators, conditions, loops, functions, dictionaries, regular expressions, error handling
Data analysis	Reading and writing data; summarizing with describe() and groupby(); filtering, cleaning and merging data
Visualization	Bar, line, scatter, histogram and pie charts; multi-chart dashboards in Matplotlib
Machine learning	Train/test split; Logistic Regression, KNN, Decision Tree, Random Forest, K-Means; accuracy and confusion matrix
Deep learning	ANN structure, activation functions, loss functions, ANN regression, binary and multi-class classification
Generative AI	LLM basics, prompt engineering, structured JSON output, using an LLM to choose tools
AI agents	Agent loop, tool registry, rule-based and LLM brains, fallback guardrail, memory
Tools	Jupyter Notebook, Google Colab, VS Code, Git and GitHub

11.2 Professional Skills Gained

- Time management: completing a defined task every day while balancing college academics.
- Communication: explaining technical ideas clearly in mentor discussions, review presentations and this report.
- Accepting feedback: incorporating mentor suggestions into the work.
- Problem solving: breaking a large goal into small steps and solving them one at a time.

11.3 Performance Evaluation by the Organization

The internship performance report issued by YBI Foundation on 16 October 2026 and signed by Dr. Alok Yadav rated my work as follows. The overall observation was "Good performance throughout the internship".

Table 15: Internship performance evaluation

No.	Criterion	Rating
1	Shows interest in work and initiative (dependable)	Good
2	Produces high quality work and accepts responsibility	Good
3	Uses technical knowledge / expertise (proficiency achieved)	Excellent
4	Analyses problems effectively	Excellent
5	Communicates well and writes effectively	Excellent
6	Punctuality (on time, time management, discipline)	Excellent
7	Accepts criticism (feedback)	Very Good
8	Teamwork (cooperation, relations with others)	Very Good
9	Professional behaviour (appearance, meeting deadlines)	Excellent
10	Demonstrates creativity / originality	Very Good

11.4 Challenges and How They Were Solved

Table 16: Challenges faced and their solutions

Challenge	How it was solved
Learning many ML algorithms and libraries in a short time	Followed the same order for every model (understand, apply, implement) and practised each one on a small dataset
Choosing a project that is useful but easy to explain	Chose a basic AI agent on the well-known Iris dataset, so the focus stays on how the agent works
Making the agent understand different ways of asking	Used several keywords per tool and regular expressions to pick out numbers; added a polite fallback message
LLM answers are not always in the expected format	Asked the LLM for JSON only, checked the tool name, and fell back to the rule-based brain on any error
Using eval() for the calculator is unsafe	Used the ast module to allow only numbers and arithmetic operators
Balancing internship work with college academics	Planned daily tasks in advance and used weekends for revision

11.5 Learning Outcomes

By the end of the internship I am able to write Python programs, analyze and visualize a dataset, choose and evaluate a suitable machine learning model for a business problem, explain the working, advantages and limitations of neural networks and classification

models, use LLMs with good prompts, and combine all of these skills into a working AI agent.

CHAPTER 12 CONCLUSION AND FUTURE SCOPE

12.1 Conclusion

The four-month internship at YBI Foundation gave me a complete, step-by-step path into Artificial Intelligence and Generative AI, from Python basics to data analysis, machine learning, deep learning and large language models. The SmartData AI Agent brought these skills together. It understands simple English requests, chooses the right tool, performs data analysis and prediction, remembers the conversation and handles unclear requests safely. It passed all eleven test cases, and its best model, KNN, reached 97.4% accuracy on unseen data. The project showed me that an AI agent is not magic: it is a clear loop of perceiving, thinking, acting, observing and responding, built from the same Python and machine learning skills learned during the internship.

12.2 Future Scope

- Allow the user to upload any CSV file, so the agent can analyze any business dataset.
- Add a web interface using Streamlit so that non-technical users can use the agent in a browser.
- Use an LLM to also write the final answer in natural language and to explain the results.
- Add more tools, such as regression for numeric prediction and an ANN model.
- Save the memory to a file so the agent remembers previous sessions.
- Add Retrieval-Augmented Generation (RAG) so the agent can answer questions from documents.

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APPENDIX A: SAMPLE RUN OF THE AGENT

The listing below is the complete output of the agent for the eleven test messages of Chapter 9, run with python agent.py --offline.

```
============================================================
 SmartData AI Agent  |  brain: Rule-based brain (offline)
============================================================
I can do these things:
  1. 'summary'                  - describe the dataset
  2. 'chart'                    - draw and save a chart
  3. 'compare models'           - train 4 ML models and compare accuracy
  4. 'predict 5.1 3.5 1.4 0.2'  - predict the species of a flower
  5. 'clusters'                 - group flowers with K-Means
  6. 'calculate 12*7+3'         - do arithmetic
  7. 'history'                  - show what we discussed
Type 'exit' to quit.

You: Agent: I can do these things:
  1. 'summary'                  - describe the dataset
  2. 'chart'                    - draw and save a chart
  3. 'compare models'           - train 4 ML models and compare accuracy
  4. 'predict 5.1 3.5 1.4 0.2'  - predict the species of a flower
  5. 'clusters'                 - group flowers with K-Means
  6. 'calculate 12*7+3'         - do arithmetic
  7. 'history'                  - show what we discussed
Type 'exit' to quit.

You: Agent: The dataset has 150 rows and 4 input columns: sepal_length, sepal_width,
petal_length, petal_width.
Flowers per species: {'setosa': 50, 'versicolor': 50, 'virginica': 50}.
Average measurements (cm) per species:
            sepal_length  sepal_width  petal_length  petal_width
species
setosa              5.01         3.43          1.46         0.25
versicolor          5.94         2.77          4.26         1.33
virginica           6.59         2.97          5.55         2.03

You: Agent: Chart saved to outputs\iris_dashboard.png

You: Agent: Model comparison on the test set (38 unseen flowers):
  Logistic Regression  accuracy = 94.7%
  KNN (k=5)            accuracy = 97.4%
  Decision Tree        accuracy = 89.5%
  Random Forest        accuracy = 92.1%
Best model: KNN (k=5)

You: Agent: Using KNN (k=5), the flower with measurements (5.1, 3.5, 1.4, 0.2) is most
likely 'setosa' (confidence 100%).

You: Agent: Using KNN (k=5), the flower with measurements (6.7, 3.0, 5.2, 2.3) is most
likely 'virginica' (confidence 100%).

You: Agent: Please give 4 numbers: sepal length, sepal width, petal length, petal
width.

You: Agent: K-Means found 3 groups. How the groups match the real species:
species  setosa  versicolor  virginica
cluster
0             0          48         14
```

```
1            50           0          0
2             0           2         36

You: Agent: (45+15)*2/3 = 40.0

You: Agent: I did not understand that. Type 'help' to see what I can do.

You: Agent: Conversation so far:
1. You: help  ->  tool: show_help
2. You: summary of the dataset  ->  tool: dataset_summary
3. You: draw a chart  ->  tool: make_chart
4. You: compare models  ->  tool: compare_models
5. You: predict 5.1 3.5 1.4 0.2  ->  tool: predict_species
6. You: which species is 6.7 3.0 5.2 2.3  ->  tool: predict_species
7. You: predict this flower  ->  tool: none
8. You: find clusters  ->  tool: find_clusters
9. You: calculate (45+15)*2/3  ->  tool: calculator
10. You: what is the weather  ->  tool: none

You: Goodbye!
```

APPENDIX B: GLOSSARY

Table 17: Terms used in this report

Term	Meaning
AI agent	A program that understands a goal, decides on an action, uses tools and returns a result
Tool	A function the agent can call to do one job, such as making a chart
Brain	The part of the agent that decides which tool to use
LLM	Large Language Model, an AI model trained on large text data that generates text
Prompt	The instruction given to an LLM
Feature	An input column used by a model, such as petal length
Label / target	The answer a supervised model learns to predict, such as the species
Training / test set	Data used to train a model / unseen data used to check it
Accuracy	Percentage of test records predicted correctly
Confusion matrix	Table comparing true classes with predicted classes
Overfitting	When a model memorizes training data and performs worse on new data
Clustering	Grouping similar records without labels
ANN	Artificial Neural Network: layers of connected neurons that learn weights
Guardrail	A check that keeps an AI system safe, such as a fallback on error

ANNEXURE I: INTERNSHIP JOINING / OFFER LETTER

Attach the offer letter and the joining confirmation letter issued by YBI Foundation.

ANNEXURE II: INTERNSHIP COMPLETION CERTIFICATE

Attach the 4-Months Internship completion certificate issued by YBI Foundation (Oct 18, 2026).

ANNEXURE III: WEEKLY INTERNSHIP REPORTS

Attach the weekly internship reports (Weeks 1-18) issued by YBI Foundation.

ANNEXURE IV: INTERNSHIP PERFORMANCE REPORT

Attach the internship performance report dated 16 October 2026.